

Example 11: Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Soln: Let p be the proposition " $\sqrt{2}$ is irrational". To start a proof by contradiction, we suppose that $\neg p$ is true. Note that $\neg p$ is the statement "It is not the case that $\sqrt{2}$ is irrational," which says that $\sqrt{2}$ is rational. We will show that assuming that $\neg p$ is true leads to a contradiction.

If $\sqrt{2}$ is rational, there exist integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors (so that the fraction a/b is in lowest terms). (Here, we are using the fact that every rational number can be written in lowest terms.) Because $\sqrt{2} = a/b$, when both sides of this equation are squared, it follows that $2 = \frac{a^2}{b^2}$. Hence, $2b^2 = a^2$.

By the definition of an even integer it follows that a^2 is even. We next use the fact that if a^2 is even, a must also be even, which follows by Exercise 18. Furthermore, because a is even, by the definition of an even integer, $a = 2c$ for some integer c . Thus,

$$2b^2 = 4c^2.$$

Dividing both sides of this equation by 2 gives $b^2 = 2c^2$. By the definition of even, this means that b^2 is even. Again, using the fact that if the square of an integer is even, then the integer itself must be even, we conclude that b must be even as well.

We have now shown that the assumption of $\neg p$ leads to the equation $\sqrt{2} = a/b$, where a and b have no common factors, but both a and b are even, that is, 2 divides both a and b . Note that the statement that $\sqrt{2} = a/b$, where a and b have no common factors, means, in particular, that 2 does not divide both a and b . $\therefore$ We have reached a contradiction and $\neg p$ must be False. $\therefore \sqrt{2}$ is irrational $\square$